

SME Customer Churn Prediction

Executive Abstract • Project Progress Review

01

SITUATION

10.02%

Baseline Churn Rate

366 of 3,652 SME accounts at risk

From Reactive to Predictive

CURRENT STATE

Descriptive analytics diagnose why customers already left — reactive, costly, too late.

TARGET STATE

Automated ML model flags high-risk accounts before exit requests are filed.

Source: Out-of-sample diagnostic data (n = 3,652)

02 KEY INSIGHT What Actually Drives Churn?

X THE MYTH

Price sensitivity drives customer exits

Historical correlation analysis shows margin fluctuations and direct tariff metrics are weak independent predictors of churn.

- Margin fluctuations
- Direct tariff sensitivity
- Contract price changes

✓ THE REALITY

Operational signals predict departure

Feature importance maps from the Random Forest model reveal consumption-based and structural account indicators as dominant behavioral precursors.

- #1 12-month energy consumption
- #2 Structural forecast metric
- #3 Forecast meter rent obligations

How well does the model perform?

Three metrics determine whether the model is ready to drive business decisions.

ACCURACY



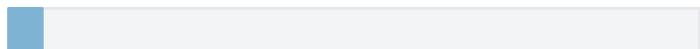
90.3% Overall correct predictions

PRECISION



78.3% True positives when flag raised

RECALL



4.9% Coverage of active churners

90.3%

Overall Accuracy

Correct identification of non-churners

78.3%

Precision

8 of 10 flagged accounts are genuine risks

0.15%

False-Positive Rate

Only 5 false alarms out of 3,286 safe accounts

What does this mean for the bottom line?

~8/10

accounts flagged are
genuine churn risks

Surgical Targeting

Retention budgets reach the right
customers — no wasted discounts on
loyal accounts.

0.15%

false-positive rate
across safe accounts

Resource Efficiency

CRM teams focus only on high-
probability saves, not chasing false
alarms.

∞

scalable automated
screening

Proactive Prevention

Real-time model scoring replaces
post-hoc analysis — prevention beats
cure.

Three Actions to Unlock Value

01

Optimize Probability Thresholds

Lower classification boundary from 0.5 default to expand recall and capture more at-risk accounts.

02

Launch Pilot Retention Campaign

Target high-probability churners from out-of-sample predictions with surgical contract interventions.

03

Integrate Engagement Data

Overlay helpdesk tickets and portal activity onto consumption profiles to sharpen future model versions.